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Roll No

ME-8004 (1) (CBGS)
B.E. VIII Semester Examination, June 2020
Choice Based Grading System (CBGS)
Optimization Technique
Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
 ii) Each question carry equal marks.

1. a) What is a design space? 7
 b) State five Engineering applications of optimization. 7
2. Determine the maximum and minimum values of the function:

$$f(x) = 12x^5 - 45x^4 + 40x^3 + 5$$
 14
3. a) What is a basic degenerate solution? 7
 b) What is the difference between a slack and surplus variable. 7
4. Solve the following LP problem by the dual simplex method: 14
 Minimize $f = 2x_1 + 9x_2 + 24x_3 + 8x_4 + 5x_5$
 Subject to: $x_1 + x_2 + 2x_3 - x_5 - x_6 = 1$
 $-2x_1 + x_3 + x_4 + x_5 - x_7 = 2$
 $x_i \geq 0, i = 1 \text{ to } 7$
5. a) What is an one-dimensional minimization problem? 7
 b) What is the difference between Fibonacci and golden section method? 7
6. Find the minimum of the function

$$f(x) = 0.64 - \frac{0.75}{1+x^2} - 0.64x \tan^{-1} \frac{1}{x}$$

 Using the following method.
 Dichotomous search method in the interval (0, 3) to achieve an accuracy of within 5% of the exact value using a value of $\delta = 0.0001$. 14
7. Define the following terms: 14
 - a) Principle of optimality
 - b) Boundary value problem
 - c) Monotonic function
 - d) Separable function

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8. Solve the following mixed integer programming problem using a graphical method:

14

Minimize $f = 4x_1 + 5x_2$

Subject to $10x_1 + x_2 \geq 10$, $5x_1 + 4x_2 \geq 20$,

$3x_1 + 7x_2 \geq 21$, $x_1 + 12x_2 \geq 12$,

$x_1 \geq 0$ and integer, $x_2 \geq 0$.
